

AS2Biz: Leveraging Web Presence and AI to Improve AS Business Classification

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Abstract

Classifying Autonomous Systems (ASes) by the business sectors of their operating organizations, also referred to as AS business classification, is a relatively recent but increasingly important area of study. However, the only publicly available dataset for this purpose suffers from significant inaccuracies and limited maintainability due to its heavy reliance on third-party sources. We argue that an organization’s web presence—particularly its official website—serves as a more reliable and transparent source for business classification, offering broad coverage and regularly updated information. We introduce AS2Biz, a web presence-based methodology for AS business classification. AS2Biz first constructs an AS-to-website mapping dataset, achieving 98.5% coverage and 94.4% accuracy, and then mainly leverages large language models to classify textual website content into business categories. The entire methodology is highly automated and cost-efficient. Our final AS2Biz dataset covers 94.3% of ASes and, on a 220-AS ground truth benchmark, achieves substantially higher accuracy—more than doubling the F1 score on subdivided business categories (0.78 vs 0.37) and improving the F1 score on general business sectors (0.87 vs 0.62).

1 Introduction

Autonomous Systems (ASes), each identified by an Autonomous System Number (ASN), form the foundational building blocks of the Internet, representing networks managed by single organizations under unified routing policies. While ASes are commonly associated with networking-focused entities such as Internet Service Providers (ISPs), hosting companies, and cloud service providers, they are also operated by a broad range of organizations across all business sectors. These include, but are not limited to, enterprises (e.g., banks and hospitals), governments, and academic institutions. Such organizations operate ASes to manage their IP address usage and establish external connectivity to the Internet.

Classifying ASes by the business sectors of their operating organizations is a relatively recent but increasingly important area of study. ASdb [35], the first and only publicly available dataset to classify ASes by business type, has been widely adopted in Internet measurement research. It has been used to identify ASes associated with specific industry sectors [27], analyze the business types of organizations involved in Internet phenomena [29, 14, 15], and uncover biases in Internet measurement platforms [30, 7].

However, in this paper we show via ground truth evaluation that ASdb suffers from significant inaccuracies, with its classifications achieving an F1 score of only 0.37 on subdivided business categories.

This is largely due to its heavy reliance on third-party databases rather than the first-hand official information—as we show in Section 2, some databases are not designed for business sector classification and others contain inaccurate information. Such reliance also introduces challenges in terms of cost and maintainability because ASdb used four paid databases (it was initially updated every six months, but at the time of writing had gone more than two years without a new release; the code is not available; and some of the commercial databases it uses are cost-prohibitive). Given the dynamic nature of the Internet ecosystem [8, 5]—with new ASes being registered regularly, frequent organizational mergers and acquisitions, and evolving business sector focus—it is important to develop an accurate, maintainable, replicable, and up-to-date AS business classification dataset.

In this paper, we introduce a novel methodology for classifying the business types of ASes based on their online presence. Specifically, we leverage their official websites as the primary information source due to several key advantages. First, broad coverage: most organizations that operate ASes maintain official websites that provide up-to-date information about their business focus, including descriptions of their services, products, and solutions. Second, compared to using third-party databases, the information from official websites is directly sourced and provides more transparency into why one or more business sectors are assigned to the AS; furthermore, websites are updated regularly. By first collecting this information and then performing classification independently, we gain greater control over the process and achieve more deterministic and accurate results compared to directly relying on pre-classified outputs from third-party databases, which often lack clarity on data provenance. Third, the entire process of collecting information from official websites and performing classification is highly automated and incurs significantly lower cost than subscribing to multiple commercial databases, enabling better maintainability and efficient regeneration of the dataset.

In a nutshell, our methodology pipeline starts with identifying the official websites of ASes, then we extract their textual information, and we prompt a state-of-the-art closed-source large language model (LLM) to classify their business sectors. For ASes lacking sufficient website information, we apply three fallback methods that rely on alternative online sources and AI services. Finally, we leverage the concept of sibling ASes (ASes controlled by the same organization) to expand our classification.

Our main contributions include: (i) AS2Web, a specialized dataset mapping ASes to their official websites, the first with a documented

and publicly reproducible methodology. It achieves 98.5% AS coverage and 94.4% accuracy. (ii) AS2Biz, a novel AS business classification methodology based on web presence, producing a dataset that covers 94.3% of ASes and, on a 220-AS ground truth benchmark, significantly outperforming ASdb across three metrics; in particular, more than doubling ASdb’s Micro-F1 on classifying subdivided business categories (0.78 vs. 0.37). (iii) Reproducibility and openness: We will open-source all relevant code for AS2Web generation, web crawling, and AI-based classification, together with the prompts used for large-language-model inference. To ensure long-term transparency and verifiable regeneration, all input webpages will be preserved in a standard web-archive format (WARC), and both the AS2Web and AS2Biz datasets will be periodically refreshed and versioned.

2 Related work and motivation

AS Classification: Dimitropoulos et al. [9] introduced the concept of AS classification in 2006, using machine learning (ML) to classify ASes into six mutually exclusive categories with different network properties and infrastructures. CAIDA AS Classification [31] was the first (2015) regularly maintained public dataset specialized in classifying ASes based on Internet features. The classification model employs a decision tree to assign each AS to one of three categories: *Transit/Access*, *Content*, or *Enterprise*, based on seven Internet-related features. Since 2021, CAIDA has ceased releasing updates to the dataset with plans to improve its accuracy.

AS Business Classification: In contrast to the traditional AS classification methods, in 2021 Ziv et al. [35] proposed an orthogonal classification concept: categorizing ASes based on the business sectors of their owning organizations. Their dataset, ASdb, is considered the state-of-the-art for AS business classification. The authors introduced a novel taxonomy, NAICSLite, a simplified version of the North American Industry Classification System (NAICS). NAICSLite adopts a two-layer hierarchical structure, comprising 17 top-level (layer-1) categories that represent general business sectors (e.g., *Computer and Information Technology*, *Finance and Insurance*) and up to 10 layer-2 categories under each top-level category, capturing subdivided business types (e.g., *ISP* under *Computer and Information Technology*). In total, NAICSLite contains 87 layer-2 categories. ASdb allows each AS to be assigned to multiple categories.

The classification approach employed by ASdb integrates multiple third-party data sources and machine learning techniques. ASdb first checks whether an ASN exists in PeeringDB [24] or IPinfo [18] and is associated with ISP-related types; if so, it classifies the ASN as *Computer and Information Technology - ISP* and terminates the classification process. For the remaining ASes, it extracts domains (with 85% ASNs coverage) from Whois data and, using additional organization attributes such as name and office address, queries a domain classification engine, Zvelo [36], as well as two firmographic databases, Dun & Bradstreet (D&B) [10] and Crunchbase [6]. In parallel, it scrapes the websites of the identified domains and feeds features extracted from such content to two separate Stochastic Gradient Descent (SGD) classifiers, each trained to perform binary classification (true/false) for one of two layer-2 categories, *ISP* and *hosting*, that fall under the *Computer and Information Technology* layer-1 category. The system then makes a final decision

comparing and ranking the output of Zvelo, D&B, Crunchbase, the *ISP* classifier, and the *hosting* classifier.

We address the same research problem but with a different approach mainly based on three steps: we first identify the official websites, we directly extract their information, and finally we engineer our own prompts to a state-of-the-art LLM to classify the business sector. We also introduce the use of sibling ASes to expand business classification (with an effect on more than 3,500 ASes in our data). In the next subsection (Section 2.1), we analyze the limitations of ASdb’s approach that contribute to classification inaccuracies and present the key insights that informed the design of our classification methodology. We evaluate the performance of AS2Biz in Section 5 by comparing AS2Biz’s earliest snapshot from January 1, 2025 to ASdb’s most recent snapshot from January 2024 (we chose these snapshots to minimize the time difference between the snapshots). For our evaluation, we apply the same taxonomy as ASdb in order to facilitate direct comparison. However, it is possible to modify AS2Biz’s taxonomy, which we discuss in Section 6.2.

Applied LLMs in Internet datasets: Chen et al. [4] proposed IPdb, an IP-level industry categorization dataset. Starting from IP addresses, the authors use reverse DNS (rDNS) queries to identify associated domains, determine the organizations operating those domains, and then gather information about the organizations from Wikipedia. They employ five LLMs to classify the Wikipedia descriptions into industry categories and train a fully connected classifier to integrate these results.

Although both IPdb and our approach leverage LLMs for business-type classification based on online information, IPdb is not an alternative solution for AS-level classification. The problem we address—classifying ASes by the business sectors of their operating organizations—differs fundamentally from IPdb’s task of labeling domains behind rDNS, especially in terms of coverage. The IP addresses classified by IPdb span only 12,318 IPv4 and 1,698 IPv6 ASes, covering roughly 16% of our classification scope (as shown in Appendix F, only 16 of the 220 ASes in our ground truth set have any IPs present in IPdb). In addition, we employ GPT-4o as the sole language model, rather than combining multiple models as IPdb does, because GPT-4o offers substantially improved text-understanding ability over GPT-3.5—the model used in IPdb—enabling accurate single-model classification.

2.1 Limitations of the state of the art

We next enumerate the three key factors that severely limit ASdb’s ability to accurately classify ASes into business sectors.

PeeringDB and IPinfo are not designed for business sector classification. Each assigns only a single category drawn from a coarse taxonomy, far less detailed than NAICSLite. By assigning the *ISP* layer-2 category to ASes solely based on relevant tags in these two sources, ASdb fails to capture the full range of business sectors represented by telecom operators that provide multiple services. For example, for China Telecom and Tata Communications, ISPs that also offer cloud and mobile phone services (NAICSLite includes layer-2 categories such as *Phone Provider* and *Hosting and Cloud*), ASdb only labels these ASes with the *ISP* subcategory.

Almost exclusive reliance on third-party databases. In ASdb, all categories are assigned exclusively through either firmographic

databases or Zvelo—a commercial domain-classification engine—except for *Computer and Information Technology*, where eight out of ten subcategories are still assigned exclusively via these sources. The two exceptions are the *ISP* and *Hosting* subcategories, where ML and (for *ISP*) PeeringDB/IPinfo are also used. In Section 1 we discussed multiple issues with third-party databases, as well as the advantages of gathering first-hand information to process thereafter. In Section 5, we empirically demonstrate that ASdb’s classification accuracy on our ground truth dataset is suboptimal across most general business sectors, highlighting inaccuracies that arise from heavy reliance on such third-party sources.

ML approaches superseded by LLMs. The ML models used by ASdb (exclusively for *ISP* and *Hosting*) to classify website text are outdated. The TF-IDF technique employed to extract textual features ignores word order and semantic meaning, limiting its ability to capture contextual information. Moreover, the SGD classifier is trained on a relatively small corpus—website text from only 225 ASes—resulting in limited generalizability. We observe numerous instances in ASdb where pure hosting providers are assigned *ISP* as one of their business categories. In contrast, modern LLMs, trained on multi-trillion-token corpora and built with advanced ML architectures, have significantly stronger capabilities in text understanding. Moreover, as generative models, LLMs can be guided through prompt design to perform a wide range of tasks. This flexibility is particularly advantageous for AS business classification under comprehensive taxonomies such as NAICS Lite, as it eliminates the need to train separate binary classifiers for each category.

3 Datasets and tools

In this section, we introduce the datasets, tools, and data service provider used in our methodology and experiments.

Scope: We classify ASNs that at a given date appear as assigned in the official delegation files of Regional Internet Registries (RIRs) and exhibited BGP activity within the preceding 12 months. To track BGP activity, we use up-to-date operational lifetime data from Nemmi et al. [21], which records the time spans during which ASes appeared in BGP paths, covering the period from 2003 to 2025. Throughout the remainder of this paper, we report results based on the snapshot taken on January 1, 2025—that is, ASes that were in an assigned status by an RIR on this date and appeared in BGP paths at any time during 2024. The total number of ASes within our classification scope is 83,808.

AS2Web Related Datasets: Whois, PeeringDB, and IPinfo are the three AS-centered datasets we use in generating our AS2Web dataset (§4.1). We use Whois databases maintained by the RIRs that store registration and contact information for each AS and its associated organization. Although their structure and policies vary across RIRs, these databases are typically organized into multiple object types linked via unique identifiers—e.g., AS objects, organization objects, and point-of-contact objects. Whois databases are maintained primarily for operational purposes, such as network coordination and abuse handling. As a result, they typically include contact email addresses for technical or abuse roles, but generally do not provide business-type or website information. However, the

domain names found in the email address fields of objects linked to ASNs can often reveal an organization’s domain.

PeeringDB [24] is a freely accessible public database in which Internet operators directly maintain information about their own ASes, including optional website URLs. Both AS objects and their associated organization objects in PeeringDB can contain website fields. However, the number of ASes with website URLs in PeeringDB is limited: as of January 2025, only 24,532 ASes include website information—accounting for approximately 29% of our classification scope.

IPinfo [18] is an online commercial platform that provides information on ASes using proprietary methods, including a website field. A total of 75,896 ASes—approximately 90% of all ASes—have associated websites in IPinfo. In this work, we utilize IPinfo’s dataset for AS domain information, obtained under their research access program.

BGP.tools [3], although not used in our AS2Web generation process, is an online platform that provides AS-related information, including websites. According to our communication with the platform’s maintainer, the website information is primarily sourced from PeeringDB and Whois, with some degree of manual verification. However, this methodology is neither peer-reviewed nor formally documented. We estimate that *BGP.tools* includes websites for approximately 50% to 70% of the ASes within our classification scope.

Sibling AS-Related Datasets: The IIL-AS2Org mapping dataset from Chen et al. [5] and the as2org+ dataset from Arturi et al. [2] are two specialized datasets for identifying sibling ASes owned by the same organization. We use them in conjunction with a manually curated sibling set covering 11 additional organizations, in the “sibling augmentation” step of our classification methodology (§4.3). In total, we identify 17,220 ASes (20.5% of our classification scope) as having at least one sibling AS.

Large Language Models: *OpenAI GPT-4o* [16] is a closed-source large language model widely regarded as a highly capable and versatile model for a broad range of tasks, offering advanced text understanding and generation capabilities. It also provides strong multilingual support, including for underrepresented languages. We confirm this performance advantage by comparing the classification results of the GPT-4o API with those of a locally deployed GPT-oss-120b model (§6). Consequently, for high-accuracy classification, we adopt GPT-4o’s chat completion API to classify both official websites and Wikipedia pages into business types (§4.2, §4.4). We use the GPT-4o snapshot dated 2024-08-06, which was the default version at the time of writing.

Perplexity AI’s Sonar-Pro [26] is a closed-source large language model optimized for real-time Internet search and summarization. It is an enhanced version of Sonar [25], an in-house model developed by Perplexity AI, based on the Llama 3.3 70B architecture. Unlike general-purpose frontier models such as GPT-4o, Sonar is designed specifically for Internet search tasks and retrieves information with inline citations from live web sources. As a more advanced variant, Sonar-Pro further improves upon Sonar. We use the Perplexity AI Sonar-Pro model at two stages in our methodology: to supplement AS2Web mappings (§4.1) and as a final fallback method to classify ASes (§4.4).

Despite Sonar-Pro’s advantage in online search, we do not adopt it as our main approach for two key reasons. First, it lacks the robustness of our main approach because we have limited control over the set of online sources that Sonar-Pro accesses and summarizes during each run. In contrast, our main approach is more deterministic, as it relies exclusively on textual content extracted from scraped websites. Second, despite prompt engineering efforts, Sonar-Pro may still include irrelevant or incorrect sources—such as references to similarly named but unrelated organizations—which can lead to inaccurate business-type classifications. In contrast, we demonstrate that the AS2Web dataset used in our main approach to identify official websites is highly accurate. Once an official website is correctly identified, GPT-4o provides more accurate and reliable classification results due to its superior text understanding capabilities compared to Sonar-Pro.

Data Service Provider: We contracted *DataWorks*, a U.S.-based data service provider, to obtain human-labeled website categories as reference data for comparing the cost and quality of website classification against GPT-4o (§4.2).

Commercial Firmographic Data Source: We use the free trial of *Crunchbase* [6] as one of the fallback methods (§4.4). *Crunchbase* is a firmographic database that supports company lookup by name or domain and returns the associated business sectors. It employs a customized classification system, organizing company profiles into over 700 predefined categories. We do not use Dun & Bradstreet [10], the other firmographic dataset used in ASdb, because we obtained a quote for accessing organizational information and found the pricing to be prohibitively high (above \$10k).

4 Classification methodology

Figure 1 provides an overview of our methodology. The first step of AS2Biz consists of mapping ASNs to the website URLs of their corresponding organization, resulting in what we call the AS2Web dataset (§4.1). For ASes with identified websites, we scrape the landing page and relevant subpages, and then classify the extracted text into NAICSlite categories using the GPT-4o API (§4.2). After completion of the website classification, we apply a sibling augmentation step to jointly consider the business focus of sibling ASes—i.e., ASes belonging to the same organization (§4.3). For each AS, we potentially assign additional business types by inheriting business categories from its sibling ASes. We include in the final AS2Biz dataset all ASes that have been assigned at least one business category—either directly from their website or inherited from sibling ASes. We refer to these first three steps as our main web-based classification approach, which classifies 85.4% of ASes. For the remaining ASes, we apply a series of fallback methods to increase AS coverage, following a prioritized order based on the reliability of each information source (§4.4). Collectively, the fallback methods contribute to the classification of an additional 8.9% of ASes.

4.1 Step 1: AS2Web mapping

Our AS2Web mapping dataset approach is outlined in Figure 2. At a high level, we leverage three AS-centered data sources—Whois, PeeringDB, and IPinfo—to identify relevant websites and supplement them with Perplexity AI queries when they do not provide

accessible websites (either not specified or specified and inaccessible). The resulting AS2Web dataset achieves 98.5% coverage (compared to 90% by IPinfo, the best among individual sources) and an extrapolated accuracy of 94.4% (compared to 84.62% by PeeringDB, 82.80% by Whois, and 81.94% by IPinfo). In the following subsections, we detail the approach, along with statistics and evaluation of the resulting dataset. More details on the evaluation are provided in Appendix B.

4.1.1 AS-centered sources. We extract domain names from all email fields associated with each AS object in Whois. We then perform extensive filtering to remove irrelevant entries, particularly those associated with public email providers. We curate a comprehensive filtering list consisting of 67 domains, which includes globally used email services, regional email providers, and default placeholder domains used by RIRs in their Whois records. Unlike Whois, both PeeringDB and IPinfo directly include website URLs. For PeeringDB, we extract the website field associated with each AS as well as its corresponding organization object.

For ASes with at least one associated domain or website URL (81,123 ASes; 96.7%), we select the most relevant domain using fuzzy string matching. We first extract and deduplicate domains from candidate URLs. For each distinct domain, we compute two similarity scores between its second-level domain (SLD) and (i) the AS name and (ii) the organization name¹. We then rank domains by the equal-weight (50/50) average of these scores and select the domain whose SLD attains the highest combined score. We then test each candidate domain over HTTPS and HTTP (with and without www). If any variant responds successfully, we record the final URL after redirects. This yields accessible website URLs for 71,530 of 81,123 ASes.

4.1.2 Perplexity AI organization website search: For the remaining 9,593 ASes with inaccessible websites, as well as the 2,685 ASes lacking website information across all three data sources, we query Perplexity AI. We first extract organization names and their corresponding registry countries from Whois data to use as search keywords in the LLM prompt when querying the Perplexity AI Sonar-Pro API. We construct a prompt with detailed instructions (see Appendix E), explicitly directing the model to use both the organization name and country to distinguish between similarly named entities across different countries. E.g., AS19076 is registered to “Weld County Government” in the U.S. Although both IPinfo and Whois list associated domains for this AS (weld.co.us and co.weld.co.us), both domains are inaccessible. Therefore, we query the organization’s website using Perplexity AI with the keywords “Weld County Government (US)”. The API returns a single website URL as the answer: <https://www.weld.gov>, which is the correct and accessible official site for this county government in Colorado, U.S.

As a result, Perplexity AI successfully identifies websites for the owning organizations of 6,426 ASes. Among these, 1,465 ASes have no website information available in any of the three data sources, while for the remaining 4,961 ASes, Perplexity AI finds a different website from the inaccessible URLs provided by AS-centered sources. We add these 6,426 ASes and their website to

¹We obtain both names from bulk WHOIS data; when unavailable, we supplement with CAIDA AS-to-Organization dataset [32].

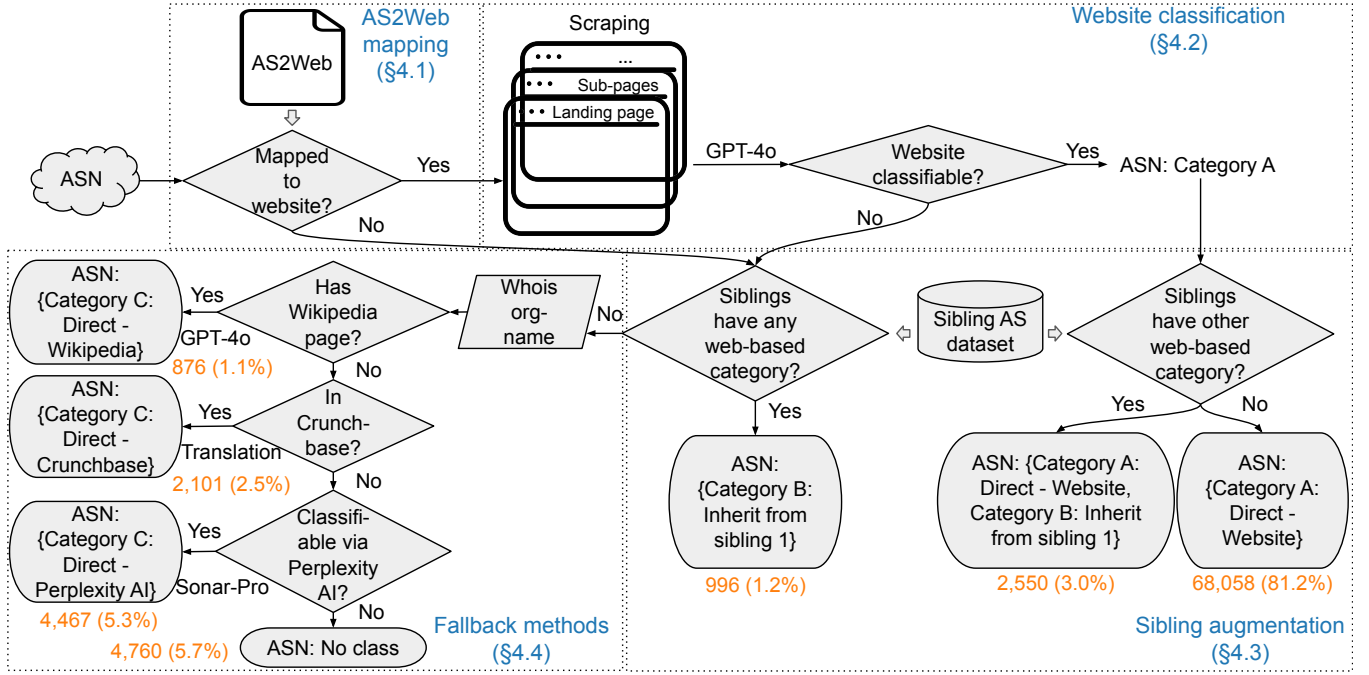


Figure 1: High-level view of our AS business classification methodology (AS2Biz). The main web-based approach consists of AS2Web mapping, website classification, and sibling augmentation, and is supplemented by a series of fallback methods. Overall AS2Biz achieves 94.3% AS coverage.

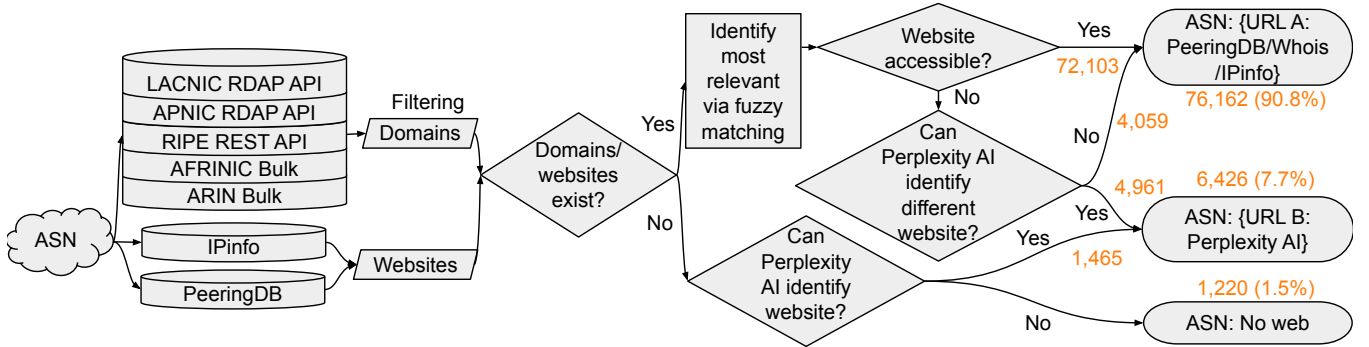


Figure 2: Pipeline of the AS2Web dataset generation approach. AS2Web achieves 98.5% of AS coverage.

our AS2Web dataset and we finally also include all the ASes with selected websites that were inaccessible, in case their unreachability is temporary².

4.1.3 Statistics and evaluation. Table 1 summarizes the statistics of our AS2Web dataset as of January 2025. The dataset provides website mappings for 82,588 ASes, covering **98.5%** of the 83,808 allocated ASes. Among these, the AS-centered data sources contribute

²When the AS-centered datasets and Perplexity AI contributed with different domains and both were inaccessible we report the one from Perplexity.

90.8% of the identified mappings, while Perplexity AI accounts for the remaining 7.7%. The individual contributions of each source to final mappings account for 72.5% (Whois), 77.1% (IPinfo), 24.8% (PeeringDB), and 7.7% (Perplexity AI) of ASes.

We evaluate our dataset by manually examining a subset of 250 sampled ASes: 192 identified by AS-centered data sources and 58 by Perplexity AI. Upon evaluation, our January 2025 AS2Web release correctly maps 236 of 250 ASes, yielding an overall accuracy of **94.4%**. The 14 errors fall into three categories: (i) Six cases originate from Whois: in three, the domain inferred from the abuse email does

Source	AS Count	#Accessible AS
All ASes	83,808 (100%)	–
AS-centered sources	76,162 (90.8%)	72,103
– Whois	59,901 (72.5%)	57,075
– IPinfo	63,713 (77.1%)	60,445
\– PeeringDB	20,483 (24.8%)	19,688
Perplexity Sonar-Pro	6,426 (7.7%)	5,750
ASes in AS2Web	82,588 (98.5%)	77,853
ASes not in AS2Web	1,220 (1.5%)	–

Table 1: Statistics of the AS2Web mapping dataset. Tree-style lines represent AS contributions from the three AS-centered sources: Whois, IPinfo, and PeeringDB. Since multiple sources may contribute the same URL for a given AS, the total contribution count exceeds the number of ASes mapped by the AS-centered sources.

not match the organization’s actual domain; in the other three, the mapped pages were under maintenance or showed default server pages, while a correct and updated website existed elsewhere. (ii) Six cases (two from IPinfo and four from Perplexity AI) map to homonymous companies located in a different country from the AS’s registered country. (iii) Two cases from Perplexity AI return a subpage of the correct organization but because it is not the main landing page, the content does not reflect the company’s business sector.

As of January 2025, RIPE’s REST API did not expose organization objects (including email fields). Consequently, for RIPE we relied on the Abuse Contact API, which contributed to the Whois abuse-email domain mismatches noted above (three cases). Following our observation and communication with RIPE, the REST API was updated in Summer 2025 to support queries for organization objects. Subsequent snapshots therefore use the organization-level email fields rather than abuse contacts. We expect that this will yield higher AS2Web accuracy than the 94.4% reported here. Since the API does not provide historical data, we cannot re-scrape websites as of January 2025, so all statistics in this paper still use the January snapshot.

4.2 Step 2: Website classification

For the 77,853 ASes with accessible websites, we proceed with scraping and content classification. While our methodology uses LLMs for content classification, we have used human annotators to collect reference data and perform evaluations as well as identify challenges in order to better develop our LLM-based approach. The lessons learned are a relevant contribution of our work and help to better frame our approach. Therefore, this section is organized as follows. We first briefly describe the scraping approach we used, we then discuss our investment in human annotation to enable this study and highlight our findings. We then provide details on our use of LLMs for website classification, and we finally compare its results against those from the human annotation, concluding that by using GPT-4o we achieve greater efficiency in both financial and temporal cost with an overall classification quality slightly better than human annotators.

4.2.1 Scraping. For each website, we capture the HTML of the landing page and a set of relevant subpages (using the playwright Python library). To identify subpages likely to describe the organization’s business, we use a curated list of 25 keywords (e.g., *about*, *business*). We detect the landing page language with langdetect and translate the keyword list using deep_translator’s GoogleTranslator. We then scan all href attributes on the landing page and follow links whose URLs contain any keyword—either in English or in the translated language—scraping their HTML as well.

4.2.2 Human annotators classification. We collaborated with three data annotators from DataWorks, beginning with training them to understand and distinguish between categories in ASdb’s NAICS-lite taxonomy. The training focused particularly on the layer-2 categories under *Computer and Information Technology*, such as *Internet Exchange Point (IXP)* and *Phone Provider*, which require more domain-specific knowledge. Upon completion of training, we adopt a labeling strategy in which two DataWorks annotators independently classify the same set of websites using the NAICS-lite taxonomy. A third annotator is then assigned to resolve any discrepancies between their labels. To support this process, we develop an online labeling interface designed to minimize time spent on repetitive tasks such as manually entering URLs, waiting for pages to load, translating non-English content, and recording business types. The interface renders the raw HTML—stored in our backend—directly within the annotators’ web interface and provides all necessary functionalities to complete the classification task efficiently. Further details about the interface design are provided in Appendix C.

As a result, the human labeling process was conducted over a period of 1.5 months, during which the three DataWorks annotators completed business type classification for 1,543 websites, at a total cost of \$5.5k. From this process, we draw a few key observations. First, hiring human annotators to classify *all* ASes is both time- and cost-prohibitive. More importantly, given our goal of periodically regenerating the dataset, such an approach is not sustainable.

Second, we observed that the existing NAICS-lite taxonomy can be imperfect, as some ASes do not fit neatly into any category that fully captures their business types. Nevertheless, we decided to continue working with and adopting NAICS-lite, mainly to be able to directly compare our approach to the state of the art. Closely related to this observation, we identified an inherent rigidity in human-led labeling: it is difficult to instruct annotators to follow even slightly modified labeling strategies for specific categories—let alone adopt an entirely revised taxonomy—once they have started labeling using the original scheme. Any such changes would require substantial re-training and re-annotation efforts. In contrast, our LLM-based classification methodology offers significant flexibility advantages. These observations are discussed further in Section 6.

4.2.3 GPT-4o classification. We adopt an LLM—specifically, GPT-4o—for the website classification task, providing as input text in the original language extracted from the landing pages and subpages of the scraped websites. To improve repeatability of model outputs across runs with identical inputs, we configure the generation parameters to reduce stochasticity. Specifically, we set *temperature* = 0, *top_p* = 1, *frequency_penalty* = 0, and *presence_penalty* = 0,

choices intended to minimize randomness in text generation. Detailed explanations of each parameter are provided in Appendix D.

Following recommended best practices for prompt engineering [23] and the prompt patterns proposed by White et al. [34], we iteratively refined our prompt design through multiple rounds. Below, we present the design of the developer prompt and user prompt we used.

Developer prompt. The developer prompt (also referred to as system prompt) provides high-level instructions to the model and is prioritized over user messages in the response generation process. In the developer prompt (see Appendix E), we first clearly define the task as classifying website textual content into a set of predefined business sectors. We then outline high-level principles that instruct the model to strictly adhere to the website content, avoiding extrapolation or guessing, which could lead to misclassification. During the prompt engineering process, we observe the model sometimes assigns categories under *Computer and Information Technology* without strictly adhering to the definitions provided in the user prompt. Additionally, we observed that GPT-4o occasionally fails to clearly distinguish between retail, manufacturing, and utility services when not explicitly guided by the prompt. To address these issues, we explicitly incorporated these clarifications into the developer prompt.

User prompt. Our user prompt provides detailed instructions for the website classification task and consists of three parts: (1) the full NAICS-lite taxonomy, (2) definitions for 12 categories that GPT-4o may misinterpret without explicit definitions, and (3) the text extracted from the website. In the first part, alongside the complete list of NAICS-lite categories, we include an additional option—*Website issue - Cannot determine categories*—to allow the LLM to label cases where the website is inaccessible or lacks sufficient informative content. This is necessary for handling cases such as default landing pages from site builders or hosting providers, as well as underdeveloped websites that display messages like “under maintenance” or “under construction”. Among the 12 specified category definitions in the second part of the user prompt, 7 belong to the *Computer and Information Technology* group—one of the most nuanced business types that requires deeper domain knowledge to distinguish accurately, which the default behavior of GPT-4o often fails to capture. The complete definitions are provided in Appendix E. In the third part, we append the text extracted from the landing page and relevant subpages, preserving the original language. To avoid exceeding the maximum input token limit, we set a threshold of 25,000 tokens per query (i.e., per website) and manually truncate the website text when necessary. In our experiments, only a small number of websites required truncation, and for the vast majority, the total token count of the complete user prompt remained below 10,000 tokens.

4.2.4 Performance Comparison Between LLM and Human Annotators. After completing the GPT-4o classification for all websites, we compare its results with those produced by human annotators. Specifically, we evaluate and compare financial cost, time efficiency, and classification quality between the two approaches.

Cost. In comparison to DataWorks, which required 1.5 months and \$5.5k to classify 1,543 websites, classifying all 68,390 distinct websites in our AS2Web dataset using GPT-4o costs approximately \$600 and can be completed within a single day. These favorable financial and temporal characteristics also make it practical to regularly regenerate and update the classification results.

Classification quality. To compare the classification results of GPT-4o and DataWorks annotators, we focus on the 1,207 websites (out of 1,543) for which both sources successfully assigned valid categories—i.e., excluding cases where either source returned no label due to insufficient website content. Since each website can be assigned multiple categories, we categorize the comparison results into four cases:

- **Case 1 (627 websites)** — The two sources have identical classifications, agreeing on both the general business sectors (layer-1) and the subdivided sectors (layer-2).
- **Case 2 (470 websites)** — The two sources agree on all layer-1 categories but differ in the layer-2 categories.
- **Case 3 (56 websites)** — The two sources share some layer-1 categories but also differ on others, resulting in partial agreement on the general business sectors.
- **Case 4 (54 websites)** — The two sources do not share any common layer-1 categories, indicating no consensus on the general business sectors.

These results indicate that GPT-4o and human annotators achieve complete agreement for 52% of websites and agree on the general business sectors for 91% of cases (Case 1 and 2). However, 48% website classifications have some discrepancies (Case 2, 3, 4).

Expert validation. To better understand the causes of these discrepancies and to assess whether GPT-4o or human annotators provide more accurate classifications, we conduct an expert validation study involving six researchers specializing in Internet measurement not involved in this work. We randomly sample 100 websites from Cases 2, 3, and 4, proportionally to their distribution across these categories. The 100 websites and six researchers are divided into three groups, each containing two researchers and a subset of 33, 33, or 34 websites. Each researcher is asked to examine the website and evaluate two *anonymized* lists of business types—one assigned by GPT-4o and one by human annotators. For each website, the researcher selects one of the following four options based on the semantic information presented on the site: *Source 1 makes more sense*, *Source 2 makes more sense*, *Both sources make sense*, *Neither source makes sense*. All researchers first perform the validation independently and then meet in their assigned two-person groups to resolve any discrepancies.

We found that LLMs and human annotators demonstrate comparable classification quality, with GPT-4o performing slightly better overall. Experts judged that for 42% of websites, the LLM-generated labels made more sense, while 39% favored the human-labeled results. The remaining 19% of cases were considered reasonable under both sources, and no instances were marked as *Neither source makes sense*.

The most common discrepancies between LLM and human annotators arise from companies offering multiple services. For example, a data center or hosting provider may also offer technical consulting and cybersecurity services, or an ISP may also provide television

services. In such cases, both sources would correctly capture at least one general business type with one source omitting other services. Both sources were similarly susceptible to this issue.

To conclude, using GPT-4o for website classification demonstrates significantly greater efficiency in both financial and temporal cost, while achieving overall classification quality slightly better than human annotators.

4.3 Step 3: Sibling augmentation

Approximately 20% of ASes within our scope are operated by organizations that own multiple ASes, where ASes under the same organization are referred to as sibling ASes. These sibling ASes may be operated by different organizational units or subsidiaries, and in some cases, may result from mergers and acquisitions. This provides two reasons for incorporating sibling relationships into the construction of our final AS2Biz dataset: (1) The business focus of one AS can be influenced by the service provisions of its sibling ASes, as parent companies often coordinate business operations across sibling ASes [5]. For example, the DDoS protection service originally offered by Prolexic (AS32787) has since been integrated into other Akamai ASes following Prolexic's acquisition by Akamai. Therefore, considering the business sectors of all sibling ASes operated by the same organization provides a more comprehensive understanding of an individual AS's business type. (2) Some ASes may lack directly associated websites due to organizational changes such as mergers, acquisitions, or renaming. In such cases, the business type of an AS can be inferred from its siblings that remain linked to the company's official website.

For the sibling augmentation step, we first curate a sibling AS dataset by combining the IIL-AS2Org [5] and as2org+ [2] datasets with a manually compiled set of sibling groups involving 11 organizations. We then construct the final AS2Biz dataset as follows: for ASes with at least one valid web-based classification (*i.e.*, not labeled as *Website issue*), we check whether any of their sibling ASes are associated with different business categories. If so, we inherit those categories and include them alongside the AS's direct web-based categories. If not, we retain and output only the AS's direct web-based categories. For ASes lacking valid categories—either due to missing website mappings or unclassifiable websites—we inherit any available valid categories from their sibling ASes. In our dataset, any inherited category is marked as such.

As a result, at the end of the sibling augmentation step, our main web-based approach successfully classifies 71,604 ASes (85.4% of all ASes). This includes 68,058 ASes with only direct categories, 996 ASes with only inherited categories, and 2,550 ASes with both direct and inherited categories.

One notable observation from the sibling augmentation step—also a special case requiring handling—involves the sibling ASes operated by China Telecom. Our manually curated sibling dataset identifies 305 ASes owned by China Telecom, many of which are regional ISP branches with organization names in Whois that include the name of a province or city but do not mention China Telecom explicitly. For these ASes, directly querying URLs based on organization names in Perplexity AI can lead to inaccurate websites. To address this, we ignore the website classification results for China Telecom ASes with Perplexity AI identified websites and

instead rely exclusively on inherited business types from sibling ASes whose websites are mapped through AS-centered sources. As a result of this special handling, AS2Biz correctly classifies all such branch ASes of China Telecom within the *Computer and Information Technology* sector by inheriting business types from China Telecom's core ASes. In contrast, ASdb misclassifies 17 of these ASes into unrelated categories such as *Service - Law, Business, and Consulting Services* and *Finance and Insurance - Insurance Carriers and Agencies*, due to its reliance on third-party databases that match on organization names. For instance, AS136186 (CHINATELECOM-Yunnan-LiJiang-MAN) is associated with the organization name "LiJiang" and it is actually a city-level Metropolitan Area Network in Yunnan Province. However, ASdb classifies this AS as *Finance and Insurance - Investment, Portfolio Management, Pensions and Funds*, likely due to directly querying the organization name in firmographic databases.

4.4 Step 4: Fallback methods

For the remaining 14.6% of ASes that cannot be classified using our main web-based approach, we implement a series of fallback methods that rely on alternative forms of online presence.

4.4.1 Wikipedia. We begin by checking whether the organization has a corresponding Wikipedia page. Following a strategy similar to IPdb [4], we extract organization names from Whois records for these ASes, derive brand names by removing common organizational suffixes, and query Wikipedia's OpenSearch API using the extracted names. If a page is returned and contains the brand name in the text, we save the full content. We then use GPT-4o—with a slightly modified developer prompt—to classify the extracted text into business categories.

4.4.2 Crunchbase. For ASes whose organizations do not have a classifiable Wikipedia page, we turn to Crunchbase, which—while not as direct a source as an official website or Wikipedia—is still more deterministic than fully relying on Perplexity AI to search for online presence. We use the organization name and, if available, the domain from our AS2Web dataset to query Crunchbase, and extract the platform's assigned industry categories. To convert Crunchbase's custom categories into our NAICSlite taxonomy, we use OpenAI's o3 model—a leading reasoning model available at the time—to generate an initial mapping from 555 involved Crunchbase categories to NAICSlite categories, which we then manually reviewed.

4.4.3 Perplexity AI. For the remaining ASes that could not be classified by any of the aforementioned methods, we use Perplexity AI to search for any available online presence of the associated organization and, based on the retrieved information, classify it into NAICSlite categories. Specifically, we use the organization name and registered country as search keywords and explicitly prompt it to (i) rely on reputable sources (*e.g.*, verified social media pages or government business directories), and (ii) ensure the organization in the cited websites has the same brand name and country or has verifiable strong connections.

Classification Stage	AS Count
All ASes	83,808 (100%)
Step 1-3: Main Web-based approach	71,604 (85.4%)
– Only direct classification	68,058 (81.2%)
– Only inherited classification	996 (1.2%)
\– Both direct and inherited	2,550 (3.0%)
Step 4: Fallback methods	7,444 (8.9%)
– Wikipedia-based classification	876 (1.1%)
– Crunchbase-based classification	2,101 (2.5%)
\– Perplexity AI-based classification	4,467 (5.3%)
ASes in AS2Biz	79,048 (94.3%)
ASes not in AS2Biz	4,760 (5.7%)

Table 2: Statistics of the AS2Biz dataset. The tree-style lines represent the breakdown of AS coverage across the main classification approach and fallback methods.

4.5 Summary

Table 2 presents the breakdown of AS classification methods and their respective coverage in our final dataset, based on the snapshot dated January 1, 2025. We first generate the AS2Web dataset, which contains 82,588 ASes—covering 98.5% of all ASes. Based on this dataset, our web-based classification approach classifies 71,604 ASes, corresponding to 85.4% of the total. The fallback methods contribute an additional 8.9% of classifications. In total, our final AS2Biz dataset achieves **94.3%** AS coverage. The total costs we incurred are \$600 for GPT-4o and \$500 for Perplexity AI. In Section 6 we discuss how in the future we plan to reduce, or ideally bring to zero, such costs.

5 Classification performance

In this section, we evaluate the classification quality of AS2Biz against ASdb³. We first compare the two datasets across all 72,069 overlapping ASes and categorize their differences, finding that AS2Biz differs from ASdb for approximately 84% of ASes. We then evaluate both datasets against a manually validated 220-AS ground truth dataset, showing that AS2Biz significantly outperforms ASdb across multiple metrics.

5.1 Large-scale comparison

Of the 79,048 ASes classified by AS2Biz, 72,069 also have labels in ASdb (the overlapping set); by AS2Biz provenance, 65,735 of these come from our web-based approach and 6,334 from fallback methods. For ASes that include only layer-1 but lack layer-2 categories in ASdb, we impute the missing layer-2 category as *Other*, following the NAICSlite taxonomy.

We then apply the four-case comparison framework introduced in Section 4.2 to categorize the overlap (Table 3). AS2Biz differs from ASdb for 83.7% of the ASes. Specifically, for 20.9% of ASes, AS2Biz assigns completely different general business sectors (Case 4); for 26.0%, there are partial disagreements within the general business sectors (Case 3); and for 36.7%, the differences lie in the subdivided

business types (Case 2). The two approaches fully agree in 16.3% of the cases (11,771 ASes).

5.2 Ground truth based evaluation

Although we have DataWorks labels, expert validation shows their quality is only comparable to GPT-4o (Section 4.2.4). To rigorously assess accuracy, we therefore construct a higher-precision ground truth dataset.

Ground truth construction. We build a manually validated ground truth dataset of 220 ASes using stratified random sampling. Stratification is based on (i) the four comparison cases in Table 3 and (ii) the coverage breakdown of our main web-based approach versus fallback methods in Table 2. As a result, we sample: 167 ASes from web-based discrepancies, 33 from web-based complete matches, 18 from fallback discrepancies, and 2 from fallback complete matches—yielding 200 ASes classified by our main web-based approach and 20 classified by fallback methods. In Appendix G, we show that this dataset’s distribution across the five RIRs largely matches the overall AS population (with APNIC slightly under-sampled and ARIN/RIPE slightly over-sampled), and that this mild skew has little effect on the results reported in this section.

To ensure quality, five researchers with expertise in Internet measurement labeled the ground truth, with each AS independently annotated by three researchers. We used the same online interface as DataWorks, augmented with more detailed instructions. For each AS, researchers: (i) verified that the website URL from our AS2Web dataset corresponds to the AS-owning organization using the provided AS name and organization name; if not, they conducted additional searches to identify the correct official website or another authoritative online source; (ii) analyzed the verified source to select business types from the NAICSlite taxonomy; (iii) when sibling ASes were shown in the interface, performed the same website verification and labeling for all siblings; and (iv) met as a three-person group to reach consensus on the final labels. Although 220 ASes constitute a relatively small sample, the labeling process is labor-intensive; we therefore prioritize annotation quality over scale.

Despite the one-year gap between our earliest AS2Biz snapshot (January 2025) and the latest ASdb release (January 2024), we believe this ground truth evaluation remains informative. We randomly sampled 20 ASes from the 220-AS set and retrieved their earliest 2024 snapshots from the Internet Archive [17]. Comparing those pages to our 2025 crawls, we found no changes that would alter business-type determinations: 18 snapshots had essentially the same layout and content, and 2 changed layout but not business sectors. While per-AS changes cannot be ruled out, this evidence suggests that one-year sector shifts are uncommon and unlikely to materially affect our results.

Metrics. We evaluate using three complementary metrics: (i) label-based Micro-F1, the standard metric for multi-label classification [33]; (ii) exact-set accuracy (strict AS-level match); and (iii) layer-1 error rate (severe misclassification into incorrect general business sectors). For Micro-F1 and exact-set accuracy, we report results at both layer-1 and layer-2 (abbreviated L1 and L2 for general and subdivided sectors, respectively). The metric definitions are as follows:

³We do not include IPinfo or IPdb as baselines because they do not aim to classify the AS-owning organization’s business sectors. See Appendix F for details.

Case Description	AS Count
Case 1: Complete match	11,771 (16.3%)
Case 2: layer-2 mismatch	26,454 (36.7%)
Case 3: Partial layer-1 mismatch	18,759 (26.0%)
Case 4: Complete layer-1 mismatch	15,085 (20.9%)
Total	72,069 (100%)

Table 3: Comparison of AS business types between AS2Biz and ASdb.

Micro-F1: Let $G_i^{(2)}$ and $P_i^{(2)}$ be the ground truth and predicted L2 label sets for AS i . Define label-level counts:

$$\begin{aligned} \text{TP}^{(2)} &= \sum_i |P_i^{(2)} \cap G_i^{(2)}|, & \text{FP}^{(2)} &= \sum_i |P_i^{(2)} \setminus G_i^{(2)}|, \\ \text{FN}^{(2)} &= \sum_i |G_i^{(2)} \setminus P_i^{(2)}|. \end{aligned}$$

Then the Micro-F1 at L2 is

$$F1_{\text{micro}}^{(2)} = \frac{2 \text{TP}^{(2)}}{2 \text{TP}^{(2)} + \text{FP}^{(2)} + \text{FN}^{(2)}}.$$

For L1, project by $G_i^{(1)} = \{L1(t) : t \in G_i^{(2)}\}$ and $P_i^{(1)} = \{L1(t) : t \in P_i^{(2)}\}$ and apply the same formula.

Exact-set accuracy: For $k \in \{1, 2\}$,

$$\text{Exact-set Accuracy}^{(k)} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{P_i^{(k)} = G_i^{(k)}\},$$

where N is the total number of ASes in the ground truth.

Cross-L1 error rate: Let $G_i^{(1)}$ be the ground truth L1 set and $P_i^{(2)}$ the predicted L2 set for AS i , and let $L1(t)$ map an L2 label t to its L1 family. We define

$$\text{Err}_{\text{cross-L1}} = \frac{\sum_i |\{t \in P_i^{(2)} : L1(t) \notin G_i^{(1)}\}|}{\sum_i |P_i^{(2)}|}.$$

Lower values indicate fewer wrong L1 predictions.

Overall performance. We evaluate all 220 ASes present in the ground truth dataset. As shown in Table 4, AS2Biz achieves substantially higher performance than ASdb across all metrics. In particular, at the layer-2 level, AS2Biz more than doubles ASdb’s performance, achieving a Micro-F1 score of 0.78 compared to ASdb’s 0.37. At the layer-1 level, the scores are 0.87 versus 0.62. Moreover, AS2Biz’s layer-1 error rate is only 6%, compared to 35% for ASdb, indicating that our approach rarely misclassifies ASes into entirely unrelated business sectors. Exact-set accuracy—which requires a complete match of all ground truth labels for an AS and serves as a strict metric for multi-label classification—yields lower absolute values due to its all-or-nothing nature, yet AS2Biz improves it by 44 *pp* at layer-2 and 33 *pp* at layer-1.

Web-based vs. fallback. The main web-based approach contributes to nearly all of the observed improvements, while fallback methods provide marginal but still measurable gains. Among the

200 ASes classified via the web-based approach, AS2Biz achieves a Micro-F1 of 0.80 (L2) and 0.90 (L1), compared to ASdb’s 0.37 (L2) and 0.64 (L1). For the remaining 20 ASes classified by fallback methods, the Micro-F1 is 0.42 (L2) and 0.64 (L1) versus 0.38 (L2) and 0.50 (L1) for ASdb. These results indicate that even for ASes with limited online presence, our fallback methods—leveraging Wikipedia and Perplexity AI—are more accurate than ASdb’s heavy reliance on third-party firmographic databases (see Appendix H). For ASes with official websites, AS2Biz provides substantially more reliable classifications on our benchmark. As shown in Figure 1, our final dataset also records the classification source for each AS, allowing users to assess the confidence level and provenance of each label.

Improvements across business sectors. Figure 3 compares AS2Biz and ASdb across all layer-1 business sectors. We highlight three observations. First, more than 60% of ASes in our ground truth (137/220; 62.3%) fall under *Computer and Information Technology*, where AS2Biz attains an F1 score higher by 0.09 than ASdb. Second, among the remaining 83 ASes (37.7%) with non-Internet business activities, AS2Biz delivers even larger improvements across all sectors—improving F1 scores by 0.17 to 1.0—except for *Health Care*. Third, the relative underperformance in *Health Care* is driven by a single case (AS27888, a Family Allowance Compensation Fund): AS2Biz correctly assigns *Finance* and *Service*, but additionally (and incorrectly) assigns *Health Care*—counting as a false positive in that sector—whereas ASdb fails to assign the correct *Finance/Service* categories but also does not assign *Health Care*, thereby avoiding a false positive in that metric. Overall, AS2Biz maintains consistently high quality across nearly all sectors and yields fewer misclassifications into incorrect layer-1 sectors, which is particularly undesirable for downstream use.

Figure 4 reports layer-2 results within *Computer and Information Technology*, the most common layer-1 sector in our dataset and, more broadly, in today’s AS ecosystem. AS2Biz achieves higher F1 across all subcategories, with especially pronounced gains for *Phone Provider*, *Hosting/Cloud Provider*, and *Technology Consulting*. In particular, we observe an overuse of the *ISP* label in ASdb (e.g., the pure hosting provider AS62005/BlueVPS is labeled as a pure *ISP*), whereas AS2Biz more accurately distinguishes among Internet-related service types.

Error analysis. We manually inspected the scraped website content and analyzed the 82 ASes where AS2Biz disagrees with the ground truth—including cases with only minor discrepancies (e.g., a difference in a single L2 category)—and identified four primary root causes: (1) AS2Web mapping errors (5 ASes; 6%), (2) web scraping coverage issues (17; 21%), (3) LLM classification errors (59; 72%), and (4) sibling augmentation (1; 1%). Two of the five AS2Web cases are corrected in subsequent snapshots after switching to RIPE organization-object email fields (as mentioned in §4.1). Among the 17 scraping-related cases, 12 result from missing navigational keywords (e.g., who-we-are) and can be resolved by expanding the keyword list, while the remaining 5 involve unpredictable or irregular href structures that are difficult to capture with rule-based crawling.

LLM classification accounts for most inaccuracies: in 33 cases the model correctly identifies the main business types but misses secondary ones despite evidence in the text; in 12 cases the labels

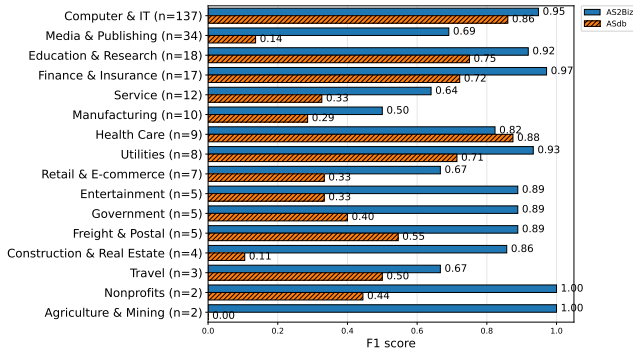


Figure 3: layer-1 classification performance (F1) against the ground truth: AS2Biz vs. ASdb. n denotes the number of ASes per sector. AS2Biz outperforms ASdb in all general sectors except *Health Care*.

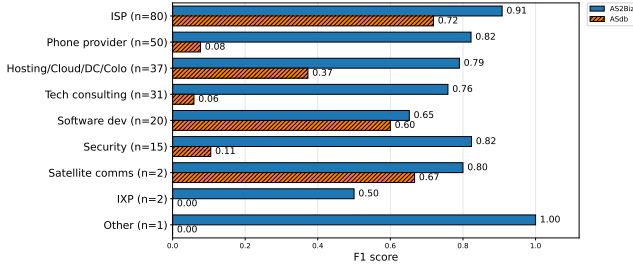


Figure 4: layer-2 classification performance (F1) under the *Computer and Information Technology* layer-1 sector: AS2Biz vs. ASdb.

include additional, inaccurate types (over-labeling); and in 14 cases the assigned types are incorrect. Finally, in one case sibling augmentation misses a business type because both sibling datasets fail to identify the relevant sibling relationship.

Overall, most inaccuracies stem from LLM classification, followed by insufficient web scraping, whereas AS2Web and sibling augmentation have only minor impact. Targeted improvements—refined prompt design and context packaging for the LLM, together with enhancements to the scraping keywords—could further improve AS2Biz’s performance.

6 Discussion and future work

6.1 Dataset sustainability

6.1.1 Local deployment of open-source LLMs: GPT-oss-120B. To further improve sustainability, we explore replacing GPT-4o in the website-classification step with an open-source model, thereby eliminating the dominant cost component in our pipeline (\$600). Specifically, we hosted GPT-oss-120B, a 120B-parameter open-source model released in August 2025 and reported to perform comparably to closed-source models on many text tasks [13], to perform website classification on a machine with two NVIDIA H100 GPUs.

Metric	ASdb	AS2Biz
layer-2 (subdivided types)		
– Micro-F1	0.37	0.78
\– Exact-set Accuracy	12%	56%
layer-1 (general sectors)		
– Micro-F1	0.62	0.87
\– Exact-set Accuracy	44%	77%
Major misclassifications		
\– layer-1 Error Rate ↓	35%	6%

Table 4: Evaluation on a 220-AS ground truth dataset. F1 scores are shown as decimals; accuracy and error rates are shown as percentages.

We evaluate GPT-oss-120B on the 200 ASes with scraped website content in our 220-AS ground truth dataset—completing classification in 55 minutes—and compare its outputs with GPT-4o and GPT-4o mini (a lightweight, lower-cost variant [12]) under identical prompts and preprocessing.

The resulting Micro-F1 scores for L2/L1 categories are as follows: GPT-4o (0.80/0.90), GPT-oss-120B (0.73/0.83), GPT-4o mini (0.62/0.78), and ASdb (0.37/0.64). In summary, the open-source model attains accuracy close to GPT-4o on both L1 and L2 categories and outperforms the lightweight closed-source GPT-4o mini.

As next steps, we plan to refine the prompts for GPT-oss-120B based on the observed errors in our analysis and incorporate techniques such as contrastive few-shot examples [20] to further guide the model toward achieving—or potentially surpassing—the current GPT-4o performance. While our pipeline is transparent and relies on publicly available LLMs, we acknowledge that the LLM component itself remains largely a black box, which can limit interpretability and systematic debugging.

6.1.2 Other sustainability-oriented efforts. Beyond local LLM deployment, we adopt several practices to ensure that each release remains efficient and reproducible. First, we locally run the method of Nemmi et al. [21] to derive operational-lifetime data, which we use to maintain an up-to-date AS scope for every snapshot. Second, for scalable website capture and storage, we design a scraper and archival pipeline around WARC—an archive format well suited to storing web content and used by the Internet Archive. Our implementation adds indexing, versioning, and content-hashing so that an identical HTML page is scraped and stored only once across snapshots (including common placeholders or CAPTCHA/challenge pages), enabling efficient long-term re-crawls. As a result, a full snapshot of ~70k websites occupies only about 10 GB of disk space.

6.1.3 Online-Search Agent with Locally Deployed LLMs. Despite our local deployment of GPT-oss-120B, supplementing AS2Web dataset and running fallback classification still require querying Perplexity AI (\$500), since vanilla locally hosted LLMs do not perform online search by default. Recent work by Alzubi et al. [1] proposes an open-source online-search agent—a controller that orchestrates a local LLM to issue searches, retrieve results, and synthesize answers.

Another direction for future work is to integrate the open-source search agent with our locally deployed GPT-oss-120B to replace Perplexity AI in both AS2Web generation and fallback classification. If effective, this will eliminate all closed-source API costs, consolidate data processing within our institutional infrastructure (improving privacy and reproducibility), and further enhance the long-term sustainability of AS2Biz.

6.2 Limitations of NAICSlite and the Flexibility of Our Methodology

From the DataWorks annotation process and expert validation study, we observe several limitations in the current NAICSlite taxonomy. For example, the layer-2 granularity within many layer-1 categories—aside from *Computer and Information Technology*—is relatively coarse, making it difficult to assign appropriate layer-2 categories for certain subdivided business types due to a lack of sufficiently specific options. For instance, under the *Agriculture, Mining, and Refineries* layer-1 category, there is only a single layer-2 option—*Farming, Greenhouses, Mining, Forestry, and Animal Farming*—which fails to distinguish among these distinct industries.

Although AS2Biz adopts the NAICSlite taxonomy, our methodology remains flexible for defining new categories and regenerating the dataset accordingly. To illustrate this flexibility, we refine the *Agriculture, Mining, and Refineries* sector by defining seven new layer-2 subcategories: *Agriculture & Farming, Mining & Minerals, Fisheries & Aquaculture, Oil & Gas, Refineries & Primary Processing, Forestry & Wood, and Other*. Using our local deployment of GPT-oss-120B, we reclassify 504 websites originally assigned to this layer-1 category by AS2Biz. The model distributes ASes across the finer-grained types, with 160, 111, 9, 112, 89, 19, and 4 ASes assigned to the respective subcategories. On a stratified 20-AS sample, we observe high layer-2 accuracy: 19 of 20 websites are correctly classified. This demonstrates that our framework can be readily extended to produce domain-specific variants of AS2Biz with minimal additional cost.

Therefore, when analyses require finer focus on specific sectors, users can leverage the AS2Web dataset to obtain official websites, apply our scraping pipeline to extract textual content or directly use our scraped website content, and adapt our LLM prompts to their research needs. In parallel, as future work we plan to design a revised taxonomy that resolves NAICSlite ambiguities and to regenerate AS2Biz accordingly.

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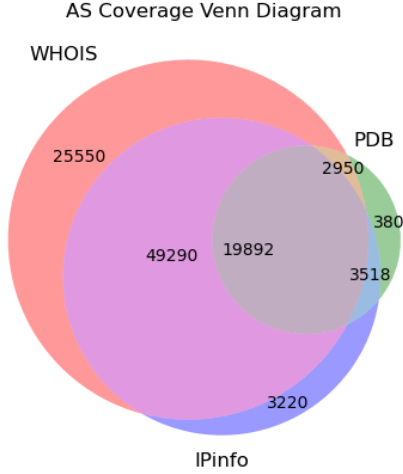


Figure 5: The AS coverage Venn diagram of Whois, PeeringDB, and IPinfo.

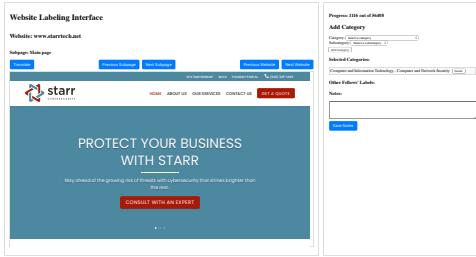


Figure 6: Screenshot of the online labeling interface we designed for DataWorks annotators.

Source	Total Checked	Correct	Incorrect or Inaccessible	Accuracy (%)
Whois	186	154	32	82.80
IPinfo	144	118	26	81.94
PeeringDB	52	44	8	84.62

Table 5: Accuracy of website mappings per data source, based on manual validation of 200 sampled ASes.

A Ethics

We submitted a protocol to our Institutional Review Board (IRB) for institutional review. The IRB determined that our study does not constitute human subjects research and therefore does not require IRB approval. Specifically, our study only involves engaging with DataWorks’s data experts who manually label the business types of a subset of organizations (and with researchers who contributed to the manual validations). We subsequently utilize their labeling results for evaluating and comparing them with a large language model (LLM). This project does not involve the collection of any personal information.

B AS2Web: individual AS-centered data source evaluation

We evaluate the effectiveness of AS-to-website mapping of the three AS-centered data sources: Whois, PeeringDB, and IPinfo. Considering only assigned ASes (i.e., without filtering by the BGP lifetime), the three sources collectively identify websites for 94.1% of ASes—104,738 out of 111,351 ASes—leaving 6,613 ASes without website information from any source. Individually, coverage is highest for Whois (87.6%), followed by IPinfo (68.2%), and PeeringDB (24%).

To evaluate the accuracy of the data sources, we manually examine the websites of 200 sampled ASes. The samples are proportionally selected based on the coverage regions in the Venn diagram (Figure 5). For example, the region where Whois and IPinfo overlap contains the largest number of ASes, from which we sample 93 ASes proportionally. The second largest region is the Whois-only region, from which we sample 49 ASes, and so on. The evaluation results for each source are summarized in Table 5. Note that an AS may appear in multiple regions (e.g., overlapping between sources), so the total number of ASes evaluated per source does not sum to exactly 200. The results indicate that all three sources exhibit comparable overall accuracy, with each achieving approximately 83%.

In addition to their similar overall accuracy, the three data sources are often complementary. For example, although PeeringDB has the highest average accuracy and is maintained directly by network operators, some entries are outdated; in such cases, Whois or IPinfo may provide more up-to-date websites. Another limitation arises in Whois: domains inferred from email fields may not match the reachable site (e.g., a top-level-domain mismatch such as .cn vs. .com). We also observe that RIPE and LACNIC Whois sometimes list email addresses belonging to a local Internet registry (LIR) that assisted with the ASN registration, rather than the AS-owning organization. In these situations, PeeringDB and IPinfo often compensate by supplying the correct domain associated with the actual AS owner.

C Web classification: labeling interface for DataWorks

To support the labeling efforts of DataWorks annotators, we build a labeling interface, aiming to minimize time spent on repetitive tasks such as manually entering URLs, waiting for websites to load, translating non-English pages, and recording business types.

Figure 6 shows a screenshot of the labeling interface. We implement the backend using the Flask library, a lightweight Python web framework. We design RESTful APIs to render the locally stored HTML of landing pages and subpages on the left panel of the frontend, and to support assigning and saving website labels on the right panel. All page-related functions are listed on the top navigation bar, with APIs enabling users to navigate through website and subpage indices. Additionally, a one-click translation feature is provided to assist annotators in understanding non-English websites. Furthermore, to ensure that annotators can access the interface remotely, we subscribe to Ngrok [22]—a tunneling service that creates secure, publicly accessible URLs for locally hosted applications—and configure a static domain for use throughout the labeling process.

D GPT-4o parameter setting

When using GPT-4o to perform the website classification task, we use the following parameter settings:

- *temperature* = 0: This disables randomness in token sampling by always selecting the most probable next token at each generation step. Setting the temperature to zero makes decoding greedy and reduces variability across runs.
- *top_p* = 1: This disables nucleus (or top-p) sampling by including the entire probability mass. When combined with *temperature* = 0, this setting has no additional effect but ensures that no stochastic filtering is applied.
- *frequency_penalty* = 0 and *presence_penalty* = 0: These settings maintain the default likelihood of repeating or introducing tokens. Since penalties for repetition can introduce variability across outputs, setting both to zero reduces potential stochasticity.

E AI prompts

We present the complete prompt used for Perplexity AI in the organizational website query during the construction of AS2Web in Box 1, and the developer prompt used for GPT-4o in our website classification step in Box 2. The detailed category definitions specified in the user prompt for all 12 business types are listed below.

- "Other - Individually Owned": "A blog or personal webpage that contains information about a specific individual. Assign this category if the website primarily showcases an individual's personal background, resume, experience, projects, or hobby infrastructure—even if technically advanced."
- "Computer and Information Technology - Internet Service Provider (ISP)": "Companies that provide individuals and organizations access to the internet via services like FTTH and FTTO. Entities operating a fiber optic backbone to deliver internet connectivity to regions or to educational and research institutions are also considered ISPs. However, if a company only constructs cellular towers or deploys fiber infrastructure without directly providing internet services, it should not be labeled as an ISP, but instead as 'Construction and Real Estate - Civil Engineering Construction (Utility Lines, Roads, and Bridges).'"
- "Computer and Information Technology - Phone Provider": "Only assign this category if the company provides direct connectivity for mobile phones (wireless), fixed-line phones, Voice over IP (VoIP), or Hosted PBX services — using its own telecommunications infrastructure or network access, and do not assign it to companies that offer only interpreting or relay services (e.g., VRS/VRI) or rely entirely on users' existing internet or cellular connections."
- "Media, Publishing, and Broadcasting - Radio and Television Providers": "Companies that provide radio and television services, including traditional broadcasting, IPTV, and TV channel offerings."
- "Computer and Information Technology - Internet Exchange Point (IXP)": "Entities that operate a physical infrastructure

where multiple networks can directly interconnect and exchange internet traffic. Networks or ASes with an open peering policy but without operating such infrastructure are not considered IXPs. If a website only mentions the network's presence at various IXPs for peering—without explicitly stating that it provides IXP services—do not assign this category."

- "Computer and Information Technology - Hosting, Cloud Provider, Data Center, Server Colocation": "Companies that provide services such as cloud computing, server hosting (including email, domain, dedicated servers, cloud servers, and VPS hosting), as well as data center and server colocation services. Do not assign this category based solely on mentions of hardware refurbishment, electronics recycling, IT asset disposition, data destruction, or data security practices, unless these are clearly part of a hosting, cloud, or colocation service offering."
- "Computer and Information Technology - Other": "Include all computer and information technology-related networks, such as experimental peering networks, IP leasing, and those that do not fit into any other computer-related categories in the above list. For experimental networks maintained by individuals out of personal interest with an open peering policy, use only this category."
- "Computer and Information Technology - Technology Consulting Services": "Companies that provide computer technology-related services, such as data analytics, technical support, solutions, and more, for specific business markets."
- "Service - Law, Business, and Consulting Services": "Companies that provide services related to law, business, or industry-specific consulting. This category excludes companies offering services in the computer or technology sectors, or those providing tech-related solutions."
- "Computer and Information Technology - Software Development": "Companies that primarily develop software products or custom software solutions. This includes web and mobile applications, enterprise software, system integration tools, and more. If a company is labeled as software development, assign additional industry-specific categories only if a particular field is clearly their primary focus. Do not assign other industry-specific categories if their products or services broadly span multiple fields without a dominant focus."
- "Retail Stores, Wholesale, and E-commerce Sites - Other": "Companies engaged in the retail, distribution, or sale of goods and equipment for specific industries—such as IT equipment, construction materials, water systems, lighting equipment, and similar. Carefully distinguish companies focused on retailing (this category) from those involved in manufacturing or providing specific services (which should be classified under other appropriate categories)."
- "Website issue - Cannot determine categories": "When a website lacks correct or sufficient information (not having enough descriptive text about its services), is under construction or maintenance, or requires login, third-party verification, or human interaction, or default webpage."

Box 1: Perplexity AI Sonar-Pro Prompt for Organization Website Search

Based on the organization’s name and the country code of its registered country, provide its official website URL: Weld County Government (US). The website must contain the exact registered name or a clear, verifiable connection to it. If multiple organizations in the same country share a similar name without a way to uniquely identify the correct one, or if no website is associated with the organization, respond with “No match.” Your response should contain only the identified URL or simply “No match.”

Box 2: Developer Prompt for OpenAI GPT-4o Chat Completion API

You are tasked with analyzing the content of a given organization’s website to classify the organization into one or more business categories. Carefully review the website’s explicit service descriptions. Only assign a category if the service is clearly and directly mentioned in the text. Do not infer or assume services based on vague, suggestive, or related language. Clearly distinguish between general business sectors—for example, companies that sell goods made by others (retail/distribution), those that manufacture such goods, and those that use them to provide services (e.g., construction or utilities). Be especially cautious when assigning subdivided categories under “Computer and Information Technology”. If there is any ambiguity, or the service is mentioned only indirectly, do not assign the category. When in doubt, recheck category definitions and default to “Other” if there’s not enough explicit evidence.

F Why we do not compare with IPdb and IPinfo

IPinfo and IPdb pursue goals that differ from ASdb and our AS2Biz, making a direct performance comparison invalid.

IPinfo. IPinfo’s ASN-type taxonomy is analogous to CAIDA’s AS classification: it characterizes Internet-facing roles rather than the business sectors of the AS-owning organization [19]. Because of the different focuses, its three mutually exclusive types—*Hosting*, *ISP*, and *Business*—also do not align with the multi-label NAICSlite business categories, precluding a meaningful evaluation against AS2Biz.

IPdb. Although IPdb assigns industry categories, it does so for IP addresses based on the hosted domains via reverse DNS, not for ASes. Because organizations commonly host their domains with third-party providers or CDNs [11], the ASes announcing those IPs are disproportionately hosting/CDN networks [4]. Moreover, IPdb does not map hosted domains to their corporate owners and then to the ASes those owners operate. Consequently, IPdb’s unit of classification and scope do not correspond to AS2Biz’s AS-level business labeling.

Empirical check: Using IPdb’s latest snapshot (June 2024) and mapping IPs to origin ASes with RouteViews Prefix2AS (2024-06-01) [28], only 16 of the 220 ASes in our ground truth set have any IPs present in IPdb. Crucially, those IP-level labels reflect the domain owners’ industries, not the business types of the hosting ASes, reinforcing the mismatch with AS2Biz.

G RIR distribution of the ground truth dataset

RIR distribution of the 220-AS evaluation set. We compare the Regional Internet Registry (RIR) composition of our 220-AS evaluation set against the overall population. In the 220-AS set, the RIR counts are: RIPE (91, 41.4%), ARIN (71, 32.3%), LACNIC (27, 12.3%), APNIC (26, 11.8%), and AFRINIC (5, 2.3%). In the overall population, the corresponding distribution is: RIPE (30,247, 38.3%), ARIN (20,581, 26.0%), APNIC (15,063, 19.1%), LACNIC (11,195, 14.2%), and AFRINIC (1,962, 2.5%). The per-RIR share differences are small for most RIRs (AFRINIC: -0.2 *pp*; LACNIC: -1.9 *pp*; RIPE: +3.1 *pp*), while APNIC is under-sampled by 7.3 *pp*, largely offset by over-sampling of ARIN (+6.3 *pp*) and RIPE (+3.1 *pp*).

To quantify any potential bias, we reweighted our aggregate metrics by the population RIR distribution. The change is as follows: Micro-F1 (L2): 0.7840 $\rightarrow$ 0.7759 ($\Delta = -0.0081$); Micro-F1 (L1): 0.8722 $\rightarrow$ 0.8641 ($\Delta = -0.0081$). The aggregate Micro-F1 scores shift by < 1 percentage point, suggesting that the mild RIR skew does not materially affect our conclusions.

H Evaluation of fallback methods

Among the 20 ASes classified by our fallback methods, 3 are labeled from Wikipedia content, 7 from Crunchbase reported types, and 10 directly via Perplexity AI. The Micro-F1 scores (L1/L2) are: Wikipedia 0.67/0.67, Crunchbase 0.57/0.26, and Perplexity AI 0.69/0.50, versus ASdb’s 0.50/0.38. These results indicate that Wikipedia and Perplexity provide stronger signal than firmographic lookups; Crunchbase improves L1 over ASdb but degrades at L2, largely due to organization/name conflation and coarse-to-NAICSlite category mappings.